

Spectral Analysis of Traffic Accidents in New York's Capital Region

Thesis Committee Presentation

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2026-04-21

Section 1

Preliminaries

Researcher

Educational Background

- M.S. Biostatistics Candidate (May 2026)
- B.S. Mathematics - University at Albany (2013)
- B.A. Political Science - University at Albany (2010)

Professional Designations

- Credentialed Property & Casualty Actuary (ACAS)
- Member, American Academy of Actuaries (MAAA)
- Chartered Property & Casualty Underwriter (CPCU)

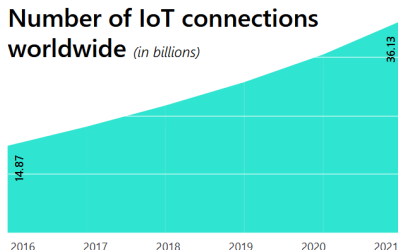
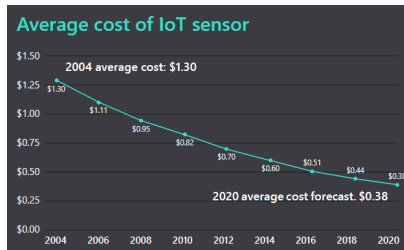
Research Question

What is the spectral density of traffic accidents in New York's Capital Region?

We evaluate the performance of the Kolmogorov-Zurbenko Fourier Transform (KZFT) in reconstructing a relative pattern in time for arrival expectation of highly random traffic accident events and validate the result out-of-time.

Growth in sensor deployment creates demand for practitioners equipped with appropriate statistical tools for analysis of signal data

- Expansive integration of sensors into consumer products
- Real-time streaming signals due to embedded connectivity (bluetooth, wi-fi)
- Reduced cost of physical data storage enabling retention of billions of data streams



Source: Microsoft 2019 Manufacturing Trends Report

While the insurance industry is rapidly incorporating sensors into many personal and commercial product lines, the actuarial profession has no educational component and almost no literature in this domain

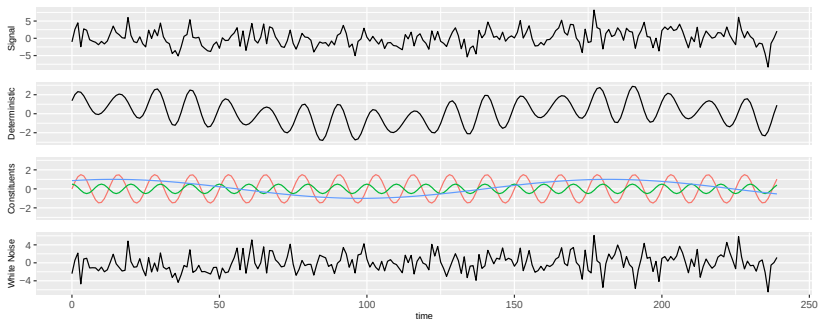
A search of the Casualty Actuarial Society's Research Library returns a single publication on spectral analysis which aims to introduce the Discrete Fourier Transform to the profession. No application is included (though in true actuarial fashion, an Excel workbook is provided).

Halliwell, L.J. (2014). "The Discrete Fourier Transform and Cyclical Overflow." *Variance* 8, 73–79, 2014.

Section 2

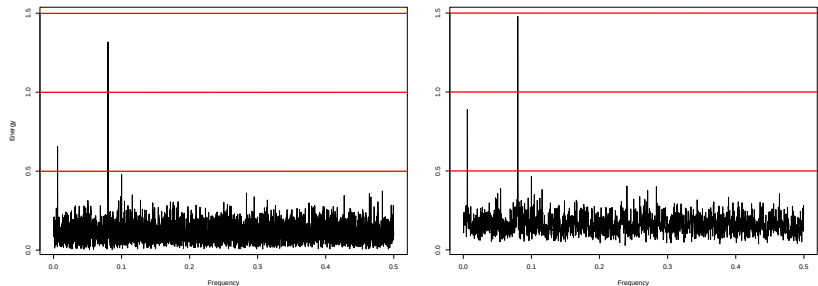
A jog to spectral density analysis with KZFT (Parts I-II)

In Parts I-II we construct a simulated signal to show the limitations of temporal methods in the context of signals governed by cyclical processes



$$X(t) = 1.5 \sin\left(\frac{2\pi}{12.5}t\right) + .5 \sin\left(\frac{2\pi}{10}t\right) + \sin\left(\frac{2\pi}{168}t\right) + N(0, \sigma = 2)$$

We review Fourier analysis and discuss estimator properties before introducing the Kolmogorov-Zurbenko Fourier Transform (KZFT), comparing their fidelity in recovering the correct energies from the signal



Spectral density (energy) based on FFT (left) and on $KZFT_{\nu,4201,6}$ to produce the KZ-periodogram (right)

Part II concludes with a discussion of the computational complexity of estimating a spectral density with KZFT, and possible solutions

- DFT (by definition): $O(n^2)$
- FFT: $O(n \log n)$
- KZFT (available implementation): $O(knm^2)$

Performance solutions

- 1 Reduce signal resolution (aggregate energy) and run multiple analyses
- 2 Evaluate only specified range of frequencies observable within window
- 3 Parallelized implementation to reduce run-time in proportion to available machine cores (*Provided in Appendix 1*)
- 4 FFT-based implementation to realize $O(knm \log m)$ complexity (*Needed*)

Section 3

Traffic Accident Analysis (Part III)

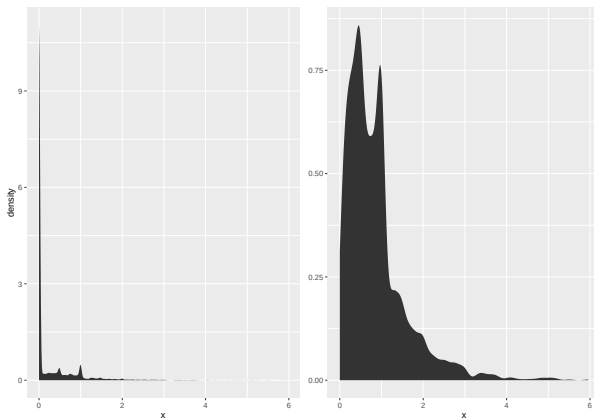
Study data provided by Moosavi, et al. (2019) contains traffic accident events with location, start and end times across the U.S. - we subset to events in the NY Capital District and take 2019 as the training period (2020-2022 will serve validation)

ID	Start_Time	End_Time	Zipcode	Description
A-2149438	2019-04-05 07:23:24	2019-04-05 08:23:08	12205	Right hand shoulder closed due
A-2149419	2019-04-05 08:17:01	2019-04-05 09:31:47	12204	Right lane blocked due to acci

The source data is restructured as a time series, with “energy” x = event duration (hours) allocated to the periods spanned

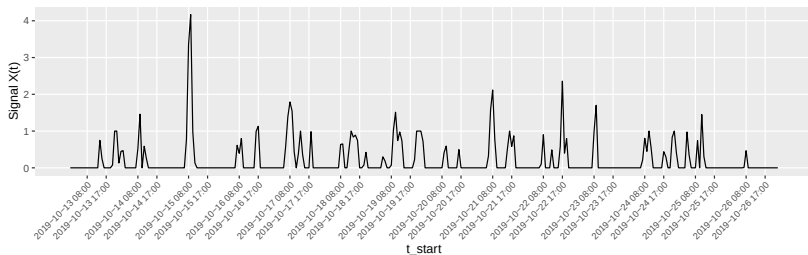
t	t_start	t_end	x
26575	2019-04-05 06:00:00	2019-04-05 07:00:00	0.0000000
26576	2019-04-05 07:00:00	2019-04-05 08:00:00	0.6100000
26577	2019-04-05 08:00:00	2019-04-05 09:00:00	1.1019444
26578	2019-04-05 09:00:00	2019-04-05 10:00:00	0.5297222
26579	2019-04-05 10:00:00	2019-04-05 11:00:00	0.0000000

Density analysis of the hourly energies shows a high variance signal with 0 the most frequent recorded energy



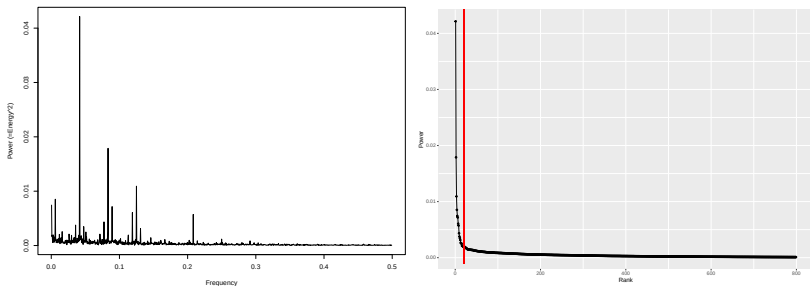
Smoothed distribution of measured hourly energy shown in total (left) and with the point-mass at 0 removed (right).

Any systematic component which governs this signal cannot be elicited by visualizing time-segments of the raw signal



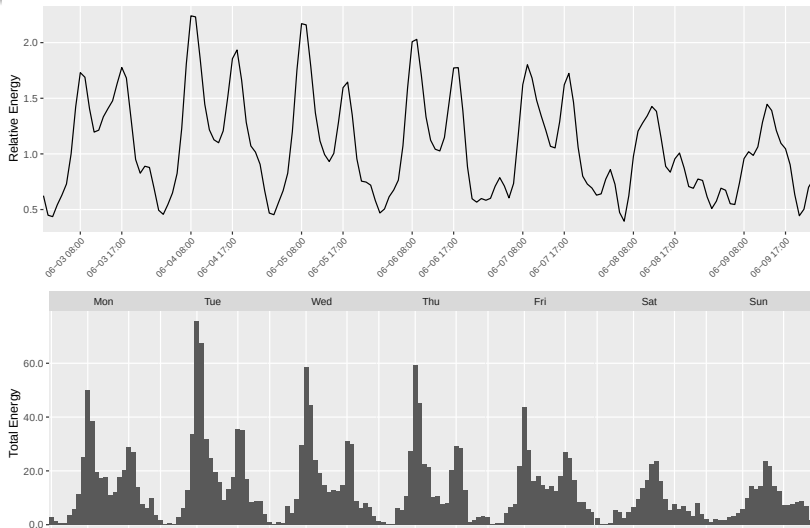
Signal energy for two-week period from Sunday, October 13 through Saturday, October 26, 2019

Power spectral density is estimated with $KZFT_{\nu,2185,3}$ revealing strong weekly pattern composed of $f_{\frac{1}{168}}$ and multiple harmonic constituents



(left) Power Spectral Density of hourly traffic accident signal; (right) scree plot of spectral constituents on power scale

Signal reconstruction based on 14 constituents produces an intuitive weekly pattern yet seems to understate the magnitudes of relative hourly energies

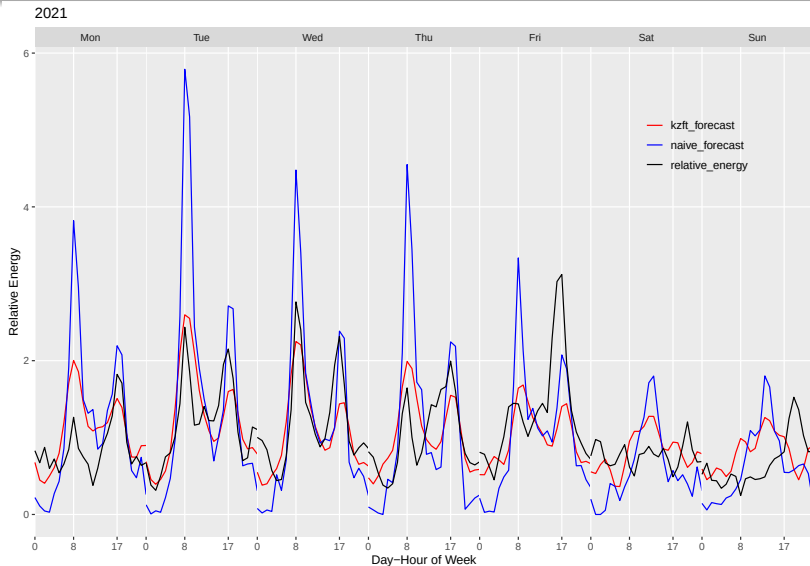


We define 2019 aggregate week-hour relative energy as our null model and compare performance to the KZFT reconstructed weekly pattern also stated relative to mean (reconstructed) energy. We evaluate each model against the patterns of years 2020-2022 individually.

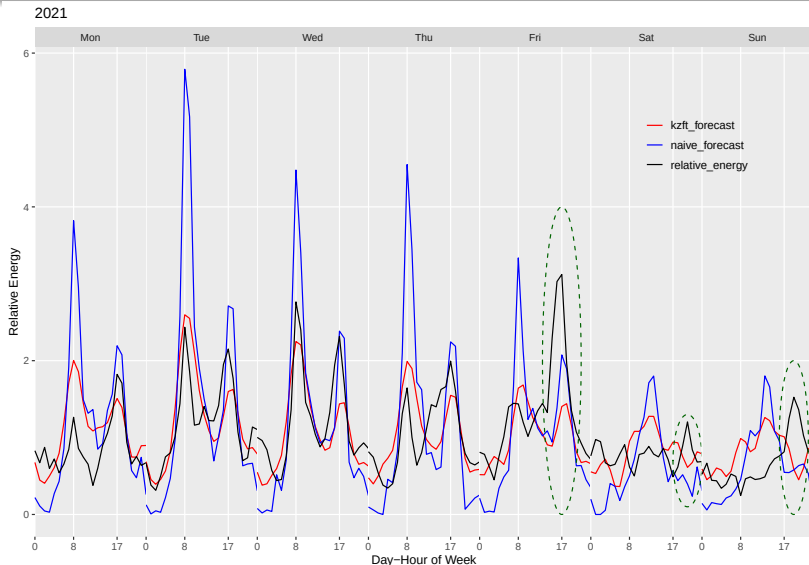
Validation result for KZFT and null model for estimated weekly pattern

Year	naive_rmse	kzft_rmse	pct_reduction
2020	1.0293	0.5813	-43.5%
2021	0.7962	0.4532	-43.1%
2022	0.8028	0.5720	-28.8%

Visual inspection shows improved (relative) forecast of KZFT.



Unexpected observation: actual results notably exceed est. Fri-Sun p.m.



This may hint at evidence of shifting driving behaviors (driver impairment) to explain increased traffic accidents and fatalities following pandemic

TNR

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Too Fast or Too Furious?

More people are dying in traffic crashes. Is pandemic rage to blame?



KZFT is suitable for analyzing non-stationary spectra over time and this is a potential area for future research.

Section 4

Conclusion

This study sought to demonstrate the applicability of spectral analysis to new domains

- Spectral analysis typically used with signals determined by physical forces - objects that spin, oscillate, vibrate, orbit
- We apply here to patterns of population behaviors
- Deployment of sensors for human activity tracking and various “smart” products presents new areas for application

Original contributions include

- Discussion of computational complexity of KZFT in comparison to FFT
- Parallel implementation of `kzp()` method for estimating power spectrum, evaluating frequencies concurrently
- Novel application to a signal not governed by physical forces
- Analysis framework which may be utilized by researchers in the insurance, auto telematics industries and beyond

Questions